

Measuring the Population Served by Community Water Systems Downstream of Coal Mining, 1990–2024: Research Question, Data, Methods, and Results

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Abstract

This note sets out and executes a measurement exercise: count the number of people whose drinking water comes from a community water system (CWS) located downstream of an active coal mine, track how that count evolved from 1990 to the present, and validate the reconstruction against held-out data. The central obstacle is that system-level population data do not exist in usable form before the mid-2000s. We therefore *reconstruct* pre-2005 population using a transparent ratio-scaling procedure that combines (i) the boundaries of each system’s service area, (ii) decennial Census population counts inside those boundaries, and (iii) the years when a system’s reported population *is* observed, used to calibrate the reconstruction. Applied to the 367 community water systems ever one HUC12 directly downstream of an active coal mine, the procedure recovers a complete 1990–2024 population series for 366 of them (99.7%). Two validation exercises — leave-one-anchor-out and leave-one-decade-out — discipline the two layers of the estimator independently and show that aggregate exposure totals are accurate even where individual systems are noisy. The document is written for readers who know applied microeconomics but not the institutional details of U.S. drinking water or the “backcasting” literature; both are introduced from scratch.

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1 Research questions

Let a *community water system* (CWS) be “downstream of coal” in year t if at least one active coal mine lies upstream of the system’s water intake in the hydrological network (Section 2 makes this precise). Write $\mathcal{D}_t$ for the set of such systems in year t , and let $S_{i,t}$ be the number of people served by system i in year t . The exercise answers four questions.

Q1. Pre-period change (1990–2005).

How did total exposed population $\sum_{i \in \mathcal{D}_t} S_{i,t}$ change between 1990 and 2005? This is the period for which $S_{i,t}$ must be reconstructed. **Answered below, Section 5.**

Q2. Recent change (2005–present).

How did the same total change between 2005 and the present, where system-level population is largely observed? **Partially answered:** the reported-served series in Section 5 runs through the anchored period; extending it to the present with the current SDWIS snapshot is future work (Section 10).

Q3. No-coal counterfactual.

If all coal mining were removed from the network, how many people would *no longer* be downstream of coal? This is a descriptive accounting counterfactual, not a causal water-quality claim. **Not yet run** — set out in Section 7 as the design for future work.

Q4. Environmental justice.

Among the exposed population, what is the demographic composition (race/ethnicity, income, age), and how does it compare to the non-exposed population? **Not yet run** — set out in Section 8 as the design for future work.

The output of the exercise is a set of *counts and distributions*, not a regression coefficient. This matters methodologically: an aggregate headcount is far more forgiving of system-level measurement error than a regression control would be, because idiosyncratic errors across systems tend to cancel in the sum — a claim the validation exercises of Section 6 test directly rather than assume.

2 Institutional background

Community water systems and PWSID. A community water system is a utility that supplies drinking water to the same resident population year-round (a municipal utility, a rural water district, a mobile-home park, etc.). Every public water system in the United States carries a

unique identifier, the PWSID, assigned in the Safe Drinking Water Information System (SDWIS), the federal regulatory database maintained by the Environmental Protection Agency (EPA). A key administrative field, `POPULATION_SERVED_COUNT`, records the estimated resident population the system serves. Two features of this field drive the entire methodology:

1. It is a *snapshot*, not a time series. EPA overwrites the value as systems file updates, so a single download yields one number per system, dated to the download, with no history.
2. It is a *reported* quantity (the population the utility claims to serve), which need not equal the residential population living inside the system’s service-area boundary — some residents use private wells, and some systems serve transient or commercial demand. The gap between the two is central below.

Watersheds and the “downstream-of-mine” classification. Surface-water flow is organized by the U.S. Geological Survey into nested watershed units; the finest standard unit used here is the twelve-digit hydrologic unit code (HUC12), a sub-watershed averaging roughly 100 km². Each HUC12 records the identity of the single HUC12 it drains into (its `tohuc`), so the HUC12 layer forms a directed flow network. A CWS is matched to the HUC12 containing its intake. Using mine locations (from the Mine Safety and Health Administration, MSHA) placed on the same network, a HUC12 — and the systems drawing from it — is classified as *downstream of a mine* in year t if an active mine sits in any HUC12 that flows into it. The target set for this exercise is narrower than that general classification: systems that are *ever*, over 1983–2024, one HUC12 *directly* downstream of an active mine, with no intake in a mining HUC12 itself. This is the same `minehuc_downstream_of_mine` classification used elsewhere in the coal-mining/drinking-water project, so the population weights built here attach to exactly the sample the project’s instrumental-variables analysis is estimated on. The annual roster $\mathcal{D}_t$ for this set is an *input* to the present exercise, not something it constructs.

Why this is hard. The roster $\mathcal{D}_t$ is known annually, but the quantity we wish to sum over it — $S_{i,t}$ — is observed only at a handful of recent dates (Section 3). The methodological core (Section 4) is a procedure to fill $S_{i,t}$ backward to 1990.

3 Data

3.1 Inputs

- **Downstream roster $\mathcal{D}_t$, 1983–2024**, annual, at the PWSID level, derived from MSHA mine activity and the HUC12 flow network. Includes system existence (entry/exit) by year. 367 systems are ever in the roster.
- **Service-area polygons** for CWS: the EPA Community Water System service-area boundary layer (SABS), the linchpin that lets us move from coarse county proxies to system-specific population (Section 4.3).
- **Census geography and demography**: tract/block boundary polygons and 100% population counts at the decennial censuses 1990, 2000, 2010, 2020 (nationwide digital boundary files begin with the 1990 TIGER/Line release, which is why the study window opens in 1990), and the Census Bureau’s Population Estimates Program (PEP) annual *county* population, used to give the system-level series an annual cadence between decennial points (Section 4.4).

3.2 Reported population-served anchors

An *anchor* is a year in which a system’s reported population served is directly observed and linkable to its PWSID. Table 1 inventories the sources used: the SYR2 occurrence databases (1998–2005), the 2006 Community Water System Survey frame, and the SDWIS 2010 and 2011 freezes. Two features of this set constrain everything downstream. First, each source supplies *one* inventory value per system, not an annual series — the SYR2 databases carry a single `POPULATION` field stamped on every sample record, so SYR2 yields one anchor per system, dated to its first sample year, not an eight-year panel. Second, no anchor of any kind exists before 1998: the 1995 and 2000 Community Water System Surveys are de-identified stratified samples that cannot be attached to a PWSID, and the 2005 SDWIS “freeze” is an 87-row summary pivot with no system records. The 1990–1997 window is therefore anchored only by extrapolation, and this is flagged throughout. The modern SDWIS snapshot is deliberately excluded from the anchor set: its `POPULATION_SERVED_COUNT` is a single current-day value broadcast across all years, the exact artifact this exercise exists to remove.

Table 1: Reported population-served anchors for downstream community water systems.

Source	Year(s)	Systems	Service-area	County	Median reported pop.
SYR2 occurrence inventory	1998–2005	140	118	22	1,646
CWSS 2006 <code>cwssframe</code>	2006	259	225	34	710
SDWIS 2010 freeze	2010	360	235	125	254
SDWIS 2011 freeze	2011	361	236	125	250
Any anchor	1998–2011	361	236	125	258
with 1 anchor		1			
with 2–3 anchors		226			
with 4 anchors		134			
No anchor (imputed)		5			

Notes: An anchor is a year in which a system’s *reported* population served is directly observed and linkable to its PWSID. Each source supplies one inventory value per system, not an annual series: the SYR2 occurrence databases carry a single `POPULATION` field stamped on every sample record, which does not vary across a system’s 1998–2005 samples, so the SYR2 anchor is dated to the system’s first sample year. The 1995 and 2000 Community Water System Surveys are de-identified samples and cannot be attached to a PWSID; the 2005 SDWIS “freeze” is a summary pivot with no system records. No anchor of any kind exists before 1998.

4 Methods

4.1 The estimand and the identity behind the reconstruction

For a system i with exposure polygon A_i , define two distinct population concepts:

$$G_{i,t} = \text{residential population living inside } A_i \text{ in year } t, \quad (\text{from Census}) \quad (1)$$

$$S_{i,t} = \text{population the system reports serving in year } t. \quad (\text{from SDWIS/anchors}) \quad (2)$$

They are linked by a system-specific *capture ratio* $r_{i,t} \equiv S_{i,t}/G_{i,t}$, which absorbs everything that makes reported service differ from polygon residence: private-well penetration, transient and commercial demand, and imprecise boundaries. The object recovered directly is $G_{i,t}$, the residential backbone, which the areal apportionment identifies without any anchoring in reported figures, and which is arguably the right notion of exposure in its own right: it counts the people living in the territory the system draws water for, whether or not the utility books them as customers.

We report *both*. Steps 1–2 (Sections 4.3–4.4) build the residential backbone $G_{i,t}$ for every system and year; Step 3 (Section 4.5) calibrates $r_{i,t}$ against observed reported population at the anchor years and forms $\hat{S}_{i,t} = \hat{r}_{i,t} \hat{G}_{i,t}$; Step 4 (Section 4.6) aggregates over the year-specific roster. The two measures bracket the truth and, as Section 5 shows, tell materially different stories about the 1990–2005 period — a divergence that is itself one of the more useful outputs of the exercise, because it isolates how much of the apparent change in exposure is an artifact of the coarser geography used for some systems.

For readers new to this style of estimate: the logic mirrors how demographers “backcast” missing population. A slowly moving, mechanically determined quantity (residents inside a fixed boundary, G) serves as a deterministic backbone, and the comparatively small, system-specific deviation from it (r) is the only thing that must be modeled.

4.2 Scope: which systems, and which polygon

Each of the 367 target systems needs a polygon into which Census population can be apportioned. Polygons are assigned in two tiers, in order of preference:

1. **Service-area polygon.** The EPA SABS layer gives the actual territory the utility serves. 239 systems match.
2. **County polygon.** For systems with no SABS boundary, the county reported in `SDWA_GEOGRAPHIC_AREAS` is used as the exposure geography. 127 systems fall into this tier, spanning 57 distinct counties.

One system matches neither and is dropped (Table 3). The tiers are not interchangeable and are kept separate in every table below: a county polygon is far larger than the service area nested inside it, so county-tier systems mechanically carry a much larger population than they truly serve. They are informative about *trends*, since a county’s growth rate is a reasonable proxy for its utilities’ growth rate, but their *levels* are upper bounds.

4.3 Step 1: apportioning Census population into the polygons

A Census block rarely coincides with a service-area boundary, so population is moved into A_i by *population-weighted areal interpolation*. For each decennial vintage $t \in \{1990, 2000, 2010, 2020\}$, intersect A_i with that vintage’s block layer and sum block population weighted by the share of each block’s area falling inside the polygon:

$$G_{i,t} = \sum_b \text{pop}_{b,t} w_{b,i,t}, \quad w_{b,i,t} = \frac{\text{area}(b \cap A_i)}{\text{area}(b)}. \quad (3)$$

Blocks are the finest Census unit and carry 100% counts at each decennial, which keeps $w_{b,i,t}$ accurate. All geometry is handled in EPSG:5070 (Albers equal area) so that the areas in (3) are undistorted.

Because the target polygon A_i is held fixed while the block layer is re-intersected at each vintage, the usual obstacle to long-run Census work — boundary redefinition every decade — does not arise: no tract-to-tract or block-to-block crosswalk is ever needed. The same overlay also yields the share of each system’s population drawn from each county, $\theta_{i,c,t_0} = G_{i,t_0}^{(c)} / G_{i,t_0}$, which feeds Step 2. Just under a quarter of service areas (57 of 239) straddle a county line, so this blend is not a formality.

4.4 Step 2: from four decennial snapshots to an annual series

Equation (3) delivers population only in Census years. To reach annual frequency, each system’s polygon population rides its county’s observed annual trajectory, taken from PEP. Let $P_{c(i),t} = \sum_c \theta_{i,c,t_0} P_{c,t}$ be the population-weighted blend of the counties the system overlaps. Starting from the most recent decennial anchor t_0 , the annual value is defined recursively by applying the county’s gross growth factor:

$$\hat{G}_{i,t} = \hat{G}_{i,t-1} \cdot \frac{P_{c(i),t}}{P_{c(i),t-1}}, \quad \hat{G}_{i,t_0} = G_{i,t_0}. \quad (4)$$

The recursion telescopes, since the cumulative product of annual growth factors collapses to the ratio of county populations at the endpoints:

$$\hat{G}_{i,t} = G_{i,t_0} \cdot \frac{P_{c(i),t}}{P_{c(i),t_0}} = \kappa_{i,t_0} P_{c(i),t}, \quad \kappa_{i,t_0} \equiv \frac{G_{i,t_0}}{P_{c(i),t_0}}. \quad (5)$$

Written this way, the content of the rule is transparent: it holds the system’s *coverage ratio* κ — the share of its county’s population that lives inside the service area — fixed at the opening decennial value, and lets the level follow the county thereafter. The single substantive assumption is that κ is constant within a decade. At each decennial the level is reset to the freshly observed apportionment and chaining resumes, so the assumption is only ever imposed over a ten-year span, never over the full window; the reset introduces a small discontinuity at decade boundaries equal to the decade’s realized drift in κ , which is the honest price of not benchmarking to the closing decennial. This is exactly the degenerate case of a full two-endpoint benchmarking rule in which the closing-decennial coverage ratio happens to equal the opening one; here it is imposed rather than estimated, and Section 6 tests the cost of that choice directly (leave-one-decade-out).

The 1990 floor is binding and cannot be pushed back: nationwide digital block boundaries begin with the 1990 TIGER/Line release. 1980 tract polygons exist only for already-tractated metropolitan areas and would introduce exactly the urban selection this exercise is trying to avoid. The roster extends back to 1983, so 1983–1989 is left unrecovered rather than extrapolated.

4.5 Step 3: the reported layer, calibrating r

The backbone above counts residents. To recover what utilities actually report serving, the capture ratio $r_{i,t} = S_{i,t}/G_{i,t}$ must be calibrated wherever S is observed and carried elsewhere.

The ratio is only meaningful for one tier. Table 2 reports the ratio’s distribution, and the split by geography tier is decisive. For service-area systems r behaves exactly as the framework assumes: it centers at 1.53, slightly above one, consistent with utilities serving some demand beyond their resident population, and the median system’s ratio moves only $1.15\times$ between its earliest and latest anchor. That drift statistic is the direct test of the constant- and interpolated- r assumptions, and it passes. For county-tier systems it does not: the denominator G is an entire county’s population, so $r \approx 0.001$ measures the utility’s share of its county rather than any capture rate. Multiplying such a ratio back by G would merely return the reported number while discarding the backbone entirely — a circular operation. The two tiers are therefore treated differently: service-area systems carry the ratio, $\hat{S}_{i,t} = \hat{r}_{i,t} \hat{G}_{i,t}$, while county-tier systems take their reported population as the *level* and the county’s growth factor as the *trend*, $\hat{S}_{i,t} = S_{i,a} (\hat{G}_{i,t} / \hat{G}_{i,a})$. Within the anchored range r is interpolated linearly between anchors and held flat outside it; the five systems with no anchor at all take the same-tier median and are flagged as least reliable.

Table 2: The capture ratio $r_{i,a} = S_{i,a}/G_{i,a}$ at anchor years, by geography tier.

	Anchors	p10	p25	Median	p75	p90
Service-area tier	814	0.842	1.068	1.530	3.169	11.530
County tier	306	0.0002	0.0005	0.0009	0.0029	0.0123
<i>Ratio drift: within-system max r / min r across anchors (service-area tier)</i>						
Systems with > 1 anchor	235	1.000	1.041	1.147	1.441	2.191

Notes: S is reported population served, G the residential population inside the system’s exposure polygon. For the service-area tier the ratio is interpretable and well behaved: it centers slightly above one, consistent with systems serving some demand beyond their resident population, and the median system’s ratio moves only $1.15\times$ between its earliest and latest anchor, which is what the constant- and interpolated- r assumptions require. For the county tier the denominator is an entire county’s population, so $r \approx 0.001$ measures the utility’s share of its county rather than a capture rate. Multiplying that ratio back by G would simply return the reported figure while discarding the backbone, so county-tier systems instead take their reported population as the level and the county’s growth as the trend.

Wholesale de-duplication. A system that sells water to another reports a population its buyer also reports, so summing both double-counts those residents. Applying the 2010 Buyers/Sellers crosswalk identifies 39 buyer–seller pairs in which both sides are downstream, implicating 27 sellers, which are removed from E_t^S before aggregation.

4.6 Step 4: aggregating over the year-specific roster

The downstream set $\mathcal{D}_t$ is known annually and changes as mines open and close. Total exposed population under each concept is

$$\text{Residential exposure: } E_t^G = \sum_{i \in \mathcal{D}_t} \hat{G}_{i,t}, \quad (6)$$

$$\text{Reported-served exposure: } E_t^S = \sum_{i \in \mathcal{D}_t} \hat{r}_{i,t} \hat{G}_{i,t}. \quad (7)$$

Summing over the year-specific roster rather than a fixed set matters: it lets the measure capture systems entering and leaving the downstream set — the extensive margin — and not only changes in population per system.

5 Results

Coverage. Table 3 reports what was recovered. Of the 367 systems ever one HUC12 directly downstream of coal mining, 366 (99.7%) receive a complete population series — 239 through their own service-area polygon and 127 through a county fallback, spanning 18 states. Only one system is lost for want of any usable geography. Each recovered system has an uninterrupted annual observation for every year from 1990 to 2024, giving a balanced panel of 12,810 system-years with no missing values.

Distribution of population. Table 4 summarizes the recovered population. The pooled distribution is strongly right-skewed, with a mean of 53,947 against a median of 3,993, and bimodal — an artifact of the two-tier geography rather than a feature of the data. Read separately, the tiers are coherent: service-area systems have a median of 480 residents against a mean of 10,234,

Table 3: Recovery of residential population for community water systems one HUC12 directly downstream of coal mining.

	Systems	Notes
<i>Panel A: Coverage of the downstream roster</i>		
Ever downstream of a mine, 1983–2024	367	Target set, $\mathcal{D} = \bigcup_t \mathcal{D}_t$
Matched to an EPA SABS service-area polygon	239	Preferred tier
Matched to a county polygon (fallback)	127	57 distinct counties
Unmatched (no polygon, no county)	1	Dropped
Population recovered	366	99.7% of the roster, across 18 states
<i>Panel B: Years recovered</i>		
Backcast window	1990–2024	35 annual observations per system
Decennial anchor years	4	1990, 2000, 2010, 2020 (block apportionment)
Intercensal years	31	PEP county growth, re-anchored each decade
System $\times$ year panel	12,810	Balanced, no missing values
Roster years not recovered	—	1983–1989: no nationwide block geography

Notes: A community water system is downstream if at least one of its intakes lies in a HUC12 directly downstream of a HUC12 with an active coal mine, and none lies in a mining HUC12 itself. Residential population inside each system’s exposure polygon is apportioned from decennial Census blocks by areal weights and carried to intercensal years with county Population Estimates Program growth. The 1990 floor binds because nationwide digital block boundaries begin with the 1990 TIGER/Line release.

consistent with the many small rural systems that dominate the downstream sample, while county-tier systems have a mean of 136,209, reflecting that they carry an entire county. The service-area figures are the ones to take at face value. The cross-sections in Panel B, which hold the set of systems fixed at all 366, show how little the underlying population moved: the median recovered system had 3,815 residents in 1990 and 4,019 in 2005.

Change from 1990 to 2005. Table 5 reports the change from the earliest recovered year to 2005, and the split between panels is the substantive result. Holding the set of systems fixed (Panel A), residential population downstream of mining is essentially flat: 19.32 million in 1990 against 19.61 million in 2005, a rise of 1.47% over fifteen years. The median system grew 0.48%, and 47.3% of systems shrank — close to an even split. The mean percentage change should be disregarded as a measure of central tendency: it is driven by a handful of very small systems growing from a near-zero base. Panel B, which sums over the year-specific roster $\mathcal{D}_t$, shows a much larger decline (residential exposure falls by more than half) because the roster itself contracts from 340 systems in 1990 to 278 in 2005.

Result: the two exposure measures diverge sharply. Table 6 sets E_t^S against E_t^G , and the contrast is the most consequential finding of the whole exercise. Reported exposure falls from 2.28 million in 1990 to 2.12 million in 2005, a decline of 7.3%. Residential exposure over the same roster falls by more than half. The ratio E_t^S/E_t^G climbs steadily from 0.159 to 0.321, the signature of a compositional shift rather than a population change.

The decomposition explains it. Of the 62 systems that leave the downstream roster between 1990 and 2005, 61 are county-tier and only one is a service area, and their median reported population is 60 people: small rural utilities that happened to lack a service-area boundary and so were assigned their whole county’s population in the residential measure. They contribute 7.92 million residents to E_{1990}^G but only 50,436 reported customers. The steep fall in E_t^G is thus substantially an artifact of the county fallback geography, not evidence that millions of people ceased to be exposed. This

Table 4: Backcast residential population of downstream community water systems: summary statistics.

	N	Mean	SD	p10	p25	Median	p75	p90	Max
<i>Panel A: Pooled system $\times$ year observations, 1990–2024</i>									
All systems	12,810	53,947	138,869	24	166	3,993	56,755	128,487	1,340,627
Service-area tier	8,365	10,234	42,706	11	61	480	3,565	20,578	537,146
County tier	4,445	136,209	204,417	29,788	42,892	78,797	144,614	319,842	1,340,627
<i>Panel B: Cross-section of all recovered systems, selected years</i>									
1990	366	52,800	141,423	16	127	3,815	53,749	124,524	1,336,449
1995	366	53,809	141,668	17	129	3,947	53,681	125,666	1,322,172
2000	366	53,497	138,866	19	150	4,060	54,487	123,762	1,281,666
2005	366	53,575	136,666	18	151	4,019	54,330	122,290	1,236,731

Notes: The unit is the residential population inside a system’s exposure polygon ($\hat{G}_{i,t}$), not the population the utility reports serving. The distribution is right-skewed and bimodal because the two geography tiers differ in scale: service-area polygons delineate the utility’s own footprint, whereas county-tier systems inherit their entire county’s population and so overstate the population attributable to any one system. Panel B holds the set of systems fixed at all recovered systems, isolating population change from roster entry and exit.

does not overturn the direction of the result — exposure still declines on both measures, and the roster still contracts from 340 systems to 278 — but it changes the magnitude by an order of magnitude and reassigns its interpretation. E_t^S is the more credible measure of how the exposed customer base actually evolved: the number of *systems* downstream of active mining fell sharply while the number of *people* they served fell modestly, because the systems that dropped out were small.

6 Validation

Six checks discipline the pipeline, all run against live data.

First, an **external check on the overlay itself**. For the 127 county-tier systems the apportionment target is a whole county, so (3) should reproduce that county’s independently measured population. Across all 508 county-tier system-years the ratio of apportioned to PEP county population lies in $[0.987, 1.036]$ with mean 0.9997: the areal machinery is not introducing meaningful bias.

Second, **internal consistency**: the county shares θ sum to one for all 1,464 system-vintage rows; the areal weights w lie in $[0, 1]$ with no missing values; and the annual panel is exactly balanced at $366 \times 35 = 12,810$ rows with no nulls or zeros.

Third, a **plausibility check on movement**. Within decades the series moves smoothly, with level resets only at decennial boundaries, as (5) implies. Eight system-years show polygon population exceeding a single county’s population; all are service areas spanning several counties, where summing across counties legitimately exceeds any one of them.

Fourth, the **ratio-drift diagnostic** in Table 2: for the 235 service-area systems with more than one anchor, the median system’s capture ratio moves $1.15\times$ between its lowest and highest anchor, and the 90th percentile moves $2.19\times$. This measures directly how much the constant- and interpolated- r assumptions strain, rather than assuming the question away.

Fifth, **leave-one-decade-out** on the $G_{i,t}$ backbone (Table 8): each of the two interior decennials, 2000 and 2010, is predicted from each of its two neighbors by PEP-scaling (5) and compared to the true block apportionment, with no reported figure anywhere in the loop. For the service-area

Table 5: Change in backcast residential population, 1990–2005.

		Total population			Change, levels		Change, %		Share
	N	1990	2005	%	Mean	Median	Mean	Median	declining
<i>Panel A: Balanced set of recovered systems (intensive margin only)</i>									
All systems	366	19,324,766	19,608,531	1.47	775	1	40.20	0.48	47.27
Service-area tier	239	2,428,429	2,404,086	-1.00	-102	1	58.44	0.75	46.86
County tier	127	16,896,337	17,204,445	1.82	2,426	151	5.88	0.47	48.03
<i>Panel B: Exposed population over the annual roster $\mathcal{D}_t$ (intensive + extensive margin)</i>									
Downstream in 1990	340	15,125,365							
Downstream in 2005	278		7,296,393	-51.76					
Stayers	278	7,203,689	7,296,393	1.29					
Exited the roster	62	7,921,677	0	-100.00					
Entered the roster	0	0	0						

Notes: Panel A holds the set of systems fixed and so measures only within-system population growth. Percentage changes are computed per system and then averaged, so the mean is inflated by a few very small systems growing from a near-zero base; the median is the more informative central tendency. Panel B sums over the year-specific downstream roster and therefore also captures systems entering and leaving the downstream set as mines open and close.

tier — the substantive test of the κ -constant assumption, since the county tier moves with the county PEP series by construction — aggregate error across the four target $\times$ direction combinations is small, -1.0% to $+1.9\%$, while median per-system absolute error is 7–12% and 70–78% of systems land within 25%. A decade of κ drift is enough to move individual systems but not enough to bias the sum, because drift is not one-signed across systems.

Sixth, **leave-one-anchor-out** on the reported-served layer $\hat{S}_{i,t}$ (Table 9): each of three eras of reported anchors — 2010–2011, 2006, and the SYR2 era (1998–2005) — is withheld in turn and re-predicted from the remaining eras using the production estimator of Section 4.5, with folds built as whole eras rather than individual anchors because 2010 and 2011 are near-duplicates (correlation 0.998; 90.8% of systems report an identical value in both years) and a per-anchor leave-one-out would trivially predict one from the other. Service-area aggregate error ranges -21.5% to $+7.3\%$ across the three folds, with median per-system absolute error 12.5–14.8%; the hardest fold, predicting the 1998–2005 anchors from only 2006 and 2010–2011 (a gap of up to 12 years), has the largest error, as expected. County-tier aggregate error is larger in magnitude (-26.1% to $+22.6\%$) but its median per-system error is smaller (1.8–6.1%), consistent with the level-and-county-growth carry being accurate for most systems but occasionally missing a large one. As with LODO, aggregate accuracy exceeds per-system accuracy throughout — the reading the exercise was designed to support rather than assume.

7 Q3: the no-coal counterfactual (not yet run)

“If all coal disappeared” can be operationalized mechanically. Let $\mathcal{D}_t$ be the factual downstream set (mines active as observed) and $\mathcal{D}_t^0$ the set recomputed after *emptying the mine layer* from the flow network. With no active mines, no system is downstream of a mine, so $\mathcal{D}_t^0 = \emptyset$ and the relieved population is simply $R_t = \sum_{i \in \mathcal{D}_t} \hat{S}_{i,t} = E_t^S$, already computed in Section 5. Two refinements would make this more informative than a single number, and are the design for future work:

1. **Intensity weighting.** Rather than a binary downstream indicator, characterize exposure by the number of upstream mines, upstream production, or upstream coal sulfur content, and

Table 6: Reported-served exposure E_t^S against residential exposure E_t^G .

Year	E_t^S (reported)	E_t^G (residential)	Ratio	Systems
1990	2,284,077	14,395,517	0.159	313
1991	2,290,054	13,840,684	0.165	308
1992	2,297,081	11,536,390	0.199	301
1993	2,308,036	11,225,835	0.206	297
1994	2,288,118	10,174,592	0.225	290
1995	2,295,309	9,529,706	0.241	283
1996	2,302,348	9,032,545	0.255	281
1997	2,304,022	8,651,149	0.266	277
1998	2,303,948	8,403,387	0.274	271
1999	2,279,462	8,005,550	0.285	268
2000	2,235,187	7,834,808	0.285	266
2001	2,210,657	7,582,810	0.292	264
2002	2,184,763	7,286,291	0.300	260
2003	2,162,210	6,772,215	0.319	256
2004	2,139,116	6,639,707	0.322	253
2005	2,117,932	6,598,723	0.321	251
<i>Decomposition of the change, 1990–2005</i>				
Downstream in 1990	2,284,077			313
Downstream in 2005	2,117,932			251
Change	-7.27%			
Stayers	2,233,642 → 2,117,932		-5.18%	251
Exited the roster	50,436	7,921,677		62

Notes: E_t^S sums estimated reported population served over the systems downstream of an active mine in year t , after removing 27 wholesale sellers whose population is already counted by a downstream buyer. The contrast with E_t^G is the substantive result. The 62 systems that leave the downstream roster between 1990 and 2005 are almost entirely county-tier (61 of 62) with a median reported population of 60 people, so they carry a large residential figure but a small reported one. The steep decline in E_t^G is therefore largely an artifact of the county fallback geography, and E_t^S is the more credible measure of how the exposed customer base actually evolved.

report the relieved population by intensity tier.

2. **Active vs. legacy coal.** Removing only *active* mines differs from removing all coal influence, since abandoned mine lands can continue to affect water; the legacy variant would be reported as a bound.

This is descriptive accounting — a count of who is currently downstream — and is explicitly *not* a claim that water quality would improve by a measured amount.

8 Q4: environmental-justice analysis (not yet run)

For each downstream system, the same polygon weights $w_{b,i,t}$ used for population (Section 4.3) can be applied to Census tract demographic shares (race/ethnicity, median household income, age, and poverty status) to obtain the demographic composition of the population served. The planned analysis would then: (i) compare the exposed population’s composition to that of the

Table 7: Annual residential population downstream of active coal mining, E_t^G .

Year	Population	Systems	Year	Population	Systems
1990	15,125,365	340	2006	7,312,518	278
1991	14,573,742	335	2007	7,045,472	274
1992	12,272,317	328	2008	7,068,446	274
1993	11,962,006	324	2009	6,907,767	271
1994	10,909,728	317	2010	6,898,693	270
1995	10,263,840	310	2011	6,315,274	267
1996	9,764,084	308	2012	6,066,657	265
1997	9,381,764	304	2013	6,002,545	263
1998	9,132,325	298	2014	5,830,862	261
1999	8,733,603	295	2015	5,738,928	259
2000	8,535,326	293	2016	5,716,969	257
2001	8,283,427	291	2017	4,416,305	255
2002	7,986,655	287	2018	4,421,952	254
2003	7,472,219	283	2019	3,994,390	250
2004	7,338,453	280	2020	3,986,997	249
2005	7,296,393	278	2021	3,859,245	248
			2022	3,874,681	248
			2023	3,813,456	246
			2024	3,838,695	245

Notes: $E_t^G = \sum_{i \in \mathcal{D}_t} \hat{G}_{i,t}$ sums backcast residential population over the systems that are one HUC12 directly downstream of an active coal mine in year t . Per-system population is roughly flat over the period, so the decline is driven by systems leaving $\mathcal{D}_t$ as upstream mines close. Levels are inflated by county-tier systems, which each contribute a whole county’s population.

non-exposed CWS population at each Census year, testing whether disadvantaged groups are over-represented downstream of coal; (ii) track these gaps over 1990–2020 to see whether the burden’s demographic profile shifted; and (iii) summarize disproportionality with standard EJ measures (population-weighted exposure shares by group; optionally a concentration/dissimilarity index). Because demographic *composition* is far more stable and better measured than the population *level*, this is expected to be the most robust output of the exercise once run, depending least on the backcasting assumptions validated in Section 6.

9 Optional machine-learning robustness (future work)

The procedure above is deliberately transparent and, for aggregate headcounts, the validation in Section 6 suggests it is adequate. As a robustness check, the simple r rule and PEP-shape interpolation could be replaced with a supervised-learning predictor (e.g. a neural network or gradient-boosted trees) trained on the anchor years to map exogenous features (Census county/place population, demographics, system type, source water) to population served, in the spirit of the demographic backcasting literature. Any such model should deliberately *exclude* SDWA violation and enforcement variables from the feature set, so that the exposure measure is not mechanically a function of the regulatory outcomes studied elsewhere in the project. Given that the deterministic backbone already explains the bulk of the variation (Section 6), a machine-learning layer is expected to refine rather than redefine the estimates, and would be reported as a sensitivity, not

Table 8: Leave-one-decade-out validation of the $G_{i,t}$ backbone. Each interior decennial is predicted from each neighboring decennial by PEP-scaling (equation 5) and compared to the true block apportionment – no reported figure enters this test.

Target year	Direction	Tier	Systems	Aggregate error	Median APE	Share $\leq 10\%$ / $\leq 25\%$
2000	Forward	Service-area	239	+1.9%	11.7%	46.4% / 70.7%
		County	127	-0.3%	0.2%	100.0% / 100.0%
2000	Backward	Service-area	239	-0.7%	10.4%	49.8% / 70.7%
		County	127	-0.1%	0.2%	100.0% / 100.0%
2010	Forward	Service-area	239	+0.8%	10.2%	49.4% / 72.0%
		County	127	+0.1%	0.2%	100.0% / 100.0%
2010	Backward	Service-area	239	-1.0%	7.1%	60.3% / 77.8%
		County	127	+0.1%	0.1%	100.0% / 100.0%

Notes: “Forward” scales the earlier decennial’s apportionment up to the target using the county’s PEP growth; “backward” scales the later decennial down. All 366 systems have all four decennials, so there is no sample loss. County-tier systems are the whole-county population by construction, so their apportionment moves exactly with the county PEP series and this fold is a near-tautological check on the PEP data itself, not on the polygon apportionment; the service-area tier is the substantive test of the κ -constant assumption.

the headline.

10 Threats to validity and limitations

1. **Pre-1998 extrapolation.** No system-level anchor exists before SYR2, so 1990–1997 exposure is a Census-residential extrapolation with r held flat at its earliest observed value. This is reported as such and the better-anchored 1998–2005 segment is leaned on for Q1.
2. **1990 polygon floor.** Nationwide tract/block polygons start in 1990; the study cannot extend earlier without metro-only, lower-quality 1980 geography, and the roster’s 1983–1989 years are left unrecovered.
3. **Service-polygon accuracy.** If a polygon poorly delineates the true service area, G is mis-measured; the r calibration partly corrects this, and the ratio-drift diagnostic (Section 6) flags the worst cases.
4. **Capture-ratio non-stationarity.** Holding/interpolating r assumes a roughly stable captured share; annexation and consolidation violate this. Measured drift bounds the error.
5. **County-tier levels are upper bounds.** 127 of 366 recovered systems carry their entire county’s population rather than a system-specific footprint, which mechanically inflates E_t^G relative to E_t^S (Section 5) and is why the reported-served series is preferred for levels.
6. **Reporting comparability.** “Population served” is defined slightly differently across SYR2, CWSS, and the SDWIS freezes; definitional drift can masquerade as real change and is checked at overlapping years.
7. **Aggregation cuts both ways.** Headcounts are robust because errors cancel (Section 6 confirms this directly); any *system-level* statement is far less reliable than the totals.
8. **Q2, Q3, Q4 are not yet executed,** only designed (Sections 1, 7, 8).

Table 9: Leave-one-anchor-out validation of the reported-served layer $\hat{S}_{i,t}$. Each fold withholds a whole era of anchors and predicts it from the survivors using the production estimator.

Held-out era	Tier	Systems	Aggregate error	Median APE	Mean APE	Share $\leq 25\%$
2010–2011 (from SYR2 + 2006)	Service-area	227	-10.3%	14.8%	41.7%	62.1%
	County	38	-26.1%	6.1%	20.9%	78.9%
2006 (from SYR2 + 2010/2011)	Service-area	225	+7.3%	13.6%	29.1%	64.9%
	County	34	+22.6%	1.8%	17.0%	82.4%
1998–2005 (from 2006 + 2010/2011)	Service-area	118	-21.5%	12.5%	28.6%	63.6%
	County	22	-4.2%	5.9%	8.5%	90.9%

Notes: Folds hold out a whole era rather than an individual anchor because 2010 and 2011 are near-duplicates (correlation 0.998; 90.8% of systems report an identical value in both years), so a per-anchor leave-one-out would predict 2011 from 2010 with near-zero error – a vacuous test. Aggregate error is the % deviation of the summed prediction from the summed held-out truth, the estimand that matters for E_t^S . APE is the per-system absolute percentage error; mean APE is inflated by small systems off a near-zero base and is reported for completeness, not as the headline. County-tier systems carry a reported level forward on the county growth factor rather than a ratio, so their error is not comparable to the service-area tier and the two are never pooled.

Summary

The known annual roster of systems downstream of coal mining ($\mathcal{D}_t$) is combined with service-area polygons and Census data to reconstruct, system by system, the population each serves, via the identity $S = r \cdot G$: Census apportionment supplies the residential backbone G for every year back to 1990, and the recent reported anchors (SYR2, CWSS 2006, and the 2010/2011 freezes) calibrate the capture ratio r . Applied to the 367 systems ever downstream of coal mining, the procedure recovers 366 complete population series and shows that residential and reported-served exposure diverge sharply — a divergence traced to the county-fallback tier rather than a mining effect. Two independent leave-one-out validations, on the κ -constant chaining assumption and on the capture-ratio calibration, both show that aggregate exposure totals are accurate even where individual systems are noisy. The design’s honest weak points are the unanchored 1990–1997 window and the upper-bound nature of county-tier levels; its most robust output, once run, would be the demographic composition underlying Q4, which barely depends on the backcasting assumptions at all.